

This API Endpoint Plan defines all routes required for RaceDay, ensuring that organisers and participants can manage events, categories, enrolments, and results in line with the approved functional requirements. The following table outlines the API endpoints required for the RaceDay system. Each endpoint is aligned with the final ERD and supports the system’s main functions, including user authentication, event and category management, participant enrolments, race results, and performance history. The endpoints also enforce appropriate role-based access and validation to maintain the integrity of the system.

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Register a new user account as an Organiser or Participant.	Public	firstName, lastName, email, password, phoneNumber, role	201 Created - user account created; 400 Bad Request - invalid input; 409 Conflict - email already registered
POST	/api/auth/login	Authenticate a user using their email and password and create an authenticated session.	Public	email, password	200 OK - authenticated user details; 400 Bad Request - missing credentials; 401 Unauthorized - invalid credentials
GET	/api/profile	Retrieve the logged-in user's account and role-specific profile information.	Logged-in User	None	200 OK - user profile; 401 Unauthorized
PUT	/api/profile	Update the logged-in user's profile information.	Logged-in User	firstName, lastName, phoneNumber	200 OK - updated profile; 400 Bad Request; 401 Unauthorized
POST	/api/events	Create a new event and automatically associate it with the authenticated Organiser.	Event Organiser	name, description, date, location, startTime, routemap, eventType, registrationDeadline, bannerImageUrl	201 Created - event with eventID, organiser ID and event details; 400 Bad Request; 401 Unauthorized; 403 Forbidden
GET	/api/events	Retrieve available events for browsing and entry.	Logged-in User	None	200 OK - array of events; 401 Unauthorized
GET	/api/events/{id}	Retrieve detailed information for a specific event, including its route and categories.	Logged-in User	None	200 OK - event details including bannerImageUrl; 401 Unauthorized; 404 Not Found
PUT	/api/events/{id}	Update an existing event; only the Organiser who created it can modify it.	Event Organiser	name, description, date, location, startTime, routemap, eventType, registrationDeadline, bannerImageUrl	200 OK - updated event; 400 Bad Request; 401 Unauthorized; 403 Forbidden; 404 Not Found
DELETE	/api/events/{id}	Delete an existing event and its associated categories and enrolment/result records where permitted by the database relationships.	Event Organiser	None	204 No Content; 401 Unauthorized; 403 Forbidden; 404 Not Found

POST	/api/events/{eventId}/categories	Create a category or distance division for an event.	Event Organiser	categoryName, distance, categoryLevel, entryFee, maxParticipants	201 Created - category with categoryID; 400 Bad Request; 401 Unauthorized; 403 Forbidden; 404 Not Found
GET	/api/events/{eventId}/categories	Retrieve all categories available for a specific event.	Logged-in User	None	200 OK - array of categories; 401 Unauthorized; 404 Not Found
POST	/api/enrolments	Register a Participant for an event and selected category, creating the PARTICIPANT_EVENT relationship.	Participant	eventId, categoryID	201 Created - enrolment with participant_eventID; 400 Bad Request; 401 Unauthorized; 403 Forbidden; 404 Not Found; 409 Conflict - duplicate enrolment or category full
GET	/api/events/{eventId}/enrolments	Retrieve participants enrolled in a specific event, including their category and registration status.	Event Organiser	None	200 OK - array of enrolments; 401 Unauthorized; 403 Forbidden; 404 Not Found
POST	/api/results	Record the official result for a participant's event enrolment.	Event Organiser	participant_eventID, time, position, status	201 Created - result with resultID, participant_eventID, time, position, status, recordedDate; 400 Bad Request; 401 Unauthorized; 403 Forbidden; 404 Not Found; 409 Conflict - result already exists
GET	/api/results/my	Retrieve all race results belonging to the logged-in Participant.	Participant	None	200 OK - results including event, category, time, position and status; 401 Unauthorized; 403 Forbidden
GET	/api/participants/me/performance	Retrieve the logged-in Participant's performance history summary based on their recorded results.	Participant	None	200 OK - performance summary containing participantID, bestTimeRecord, totalWins, bestPositionHeld, lastRace, averagePosition; 401 Unauthorized; 403 Forbidden

Notes to API

- Role-based access: Organiser and Participant endpoints are restricted according to the user's assigned role.
- Authentication: Registration and login establish the user's authenticated session, which is used to control access to protected endpoints.
 - Event ownership: When an Organiser creates an event, it is automatically linked to that Organiser. Organisers can only update or delete their own events.
- Categories: Each event can contain multiple categories, with maxParticipants used to control category capacity.
- Enrolments: The enrolment endpoint creates the PARTICIPANT_EVENT relationship between a Participant, Event and Category.
- System-controlled data: Values such as IDs, registrationDate, registrationStatus, bib, and recordedDate are generated or controlled by the system rather than entered manually by users.
- Results: A result is linked to a specific PARTICIPANT_EVENT, allowing the system to associate race times, positions and statuses with the correct participant and event.
- Performance history: Performance history is based on recorded results and provides summary information such as bestTimeRecord, totalWins, bestPositionHeld, lastRace, and averagePosition.
- Validation and errors: The API uses appropriate response codes such as 400, 401, 403, 404, and 409 to handle invalid input, authentication, authorisation, missing records and business-rule conflicts.
 - ERD alignment: The API reflects the relationships in the final ERD, particularly the flow from Event → Category → Participant Event → Result.